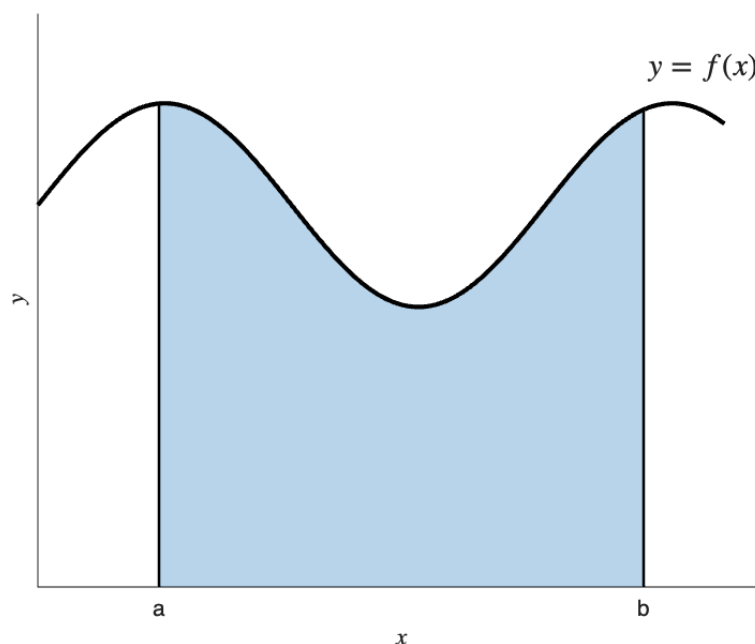


## 4.2 Integration Fundamentals

Integration is one of the two fundamental operations of calculus. Whereas differentiation determines the rate at which a quantity changes, integration determines the accumulated effect of a quantity over an interval. One important application of integration is the calculation of areas beneath curves.

### 4.2.1 Area Under a Curve

Consider the function  $y = f(x)$ . The shaded region below represents the area between the curve and the  $x$ -axis over a specified interval,  $[a, b]$ .



The area between the curve, the  $x$ -axis and the vertical lines  $x = a$  and  $x = b$  can be determined using integration.

#### Definition

Integration can be used to calculate the area under a curve between two specified values of the independent variable.

### 4.2.2 Integration as the Reverse of Differentiation

Recall that differentiation allows us to determine the derivative of a function. For example,

$$\frac{d}{dx}(x^2) = 2x$$

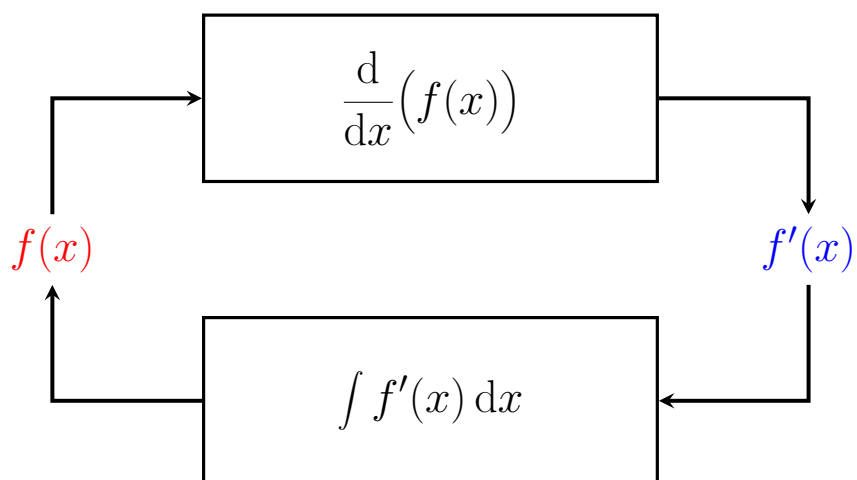
Working backwards, if

$$\frac{dy}{dx} = 2x,$$

then

$$y = x^2$$

This process of working backwards is called integration. The relationship between differentiation and integration is illustrated below.



### 4.2.3 Integration Notation

The symbol used to represent integration is

$$\int$$

which is known as the **integral sign**.

### Definition

An expression of the form

$$\int f(x) \, dx$$

means:

“Find a function whose derivative is  $f(x)$ .”

The symbol

$$dx$$

indicates that integration is being performed with respect to  $x$ .

For example,

$$\int 2x \, dx$$

asks us to find a function whose derivative is  $2x$ . Since

$$\frac{d}{dx}(x^2) = 2x,$$

it follows that

$$\int 2x \, dx = x^2$$

However, this answer is incomplete.

### 4.2.4 The Constant of Integration

The derivative of a constant is always zero. For example,

$$\frac{d}{dx}(x^2 + 3) = 2x$$

and

$$\frac{d}{dx}(x^2 - 8) = 2x$$

Therefore many different functions possess the same derivative.

### Definition

Every indefinite integral must include a constant

$$+C$$

called the **constant of integration**.

Therefore

$$\int 2x \, dx = x^2 + C$$

## 4.2.5 Standard Integration Rules

The following integration rules appear in the formulae booklet.

### Definition

For constants  $k$ ,  $a$ ,  $b$  and  $n$ :

| Function      | Integral                                     |
|---------------|----------------------------------------------|
| $k$           | $kx + C$                                     |
| $ax^n$        | $\frac{ax^{n+1}}{n+1} + C \quad (n \neq -1)$ |
| $a e^{bx}$    | $\frac{a}{b} e^{bx} + C$                     |
| $\frac{a}{x}$ | $a \ln(x) + C$                               |
| $a \cos(bx)$  | $\frac{a}{b} \sin(bx) + C$                   |
| $a \sin(bx)$  | $-\frac{a}{b} \cos(bx) + C$                  |

The integral operator  $\int dx$  is a **linear operator**. This means:

1. Coefficients may be taken outside the operator (this has already been covered within the standard formulae)
2. The operator may be applied term-by-term.

## 4.2.6 Indefinite Integration

An integral that contains no limits is called an indefinite integral.

### Definition

An **indefinite integral** produces a general expression containing the constant of integration  $C$ .

### Worked Example

Evaluate

$$\int 6x^4 \, dx$$

Using the power rule:

$$\begin{aligned}\int 6x^4 \, dx &= \frac{6x^{4+1}}{4+1} + C \\ &= \frac{6}{5}x^5 + C\end{aligned}$$

### Worked Example

Evaluate

$$\int 3 \cos(8x) + 4 \, dx$$

Integrating term-by-term:

$$\begin{aligned}\int 3 \cos(8x) + 4 \, dx &= \int 3 \cos(8x) \, dx + \int 4 \, dx \\ &= \frac{3}{8} \sin(8x) + 4x + C\end{aligned}$$

### Worked Example

Evaluate

$$\int 7e^{-3x} - 5 \, dx$$

$$\begin{aligned}\int 7e^{-3x} - 5 \, dx &= \frac{7}{-3} e^{-3x} - 5x + C \\ &= -\frac{7}{3} e^{-3x} - 5x + C\end{aligned}$$

## 4.2.7 Definite Integration

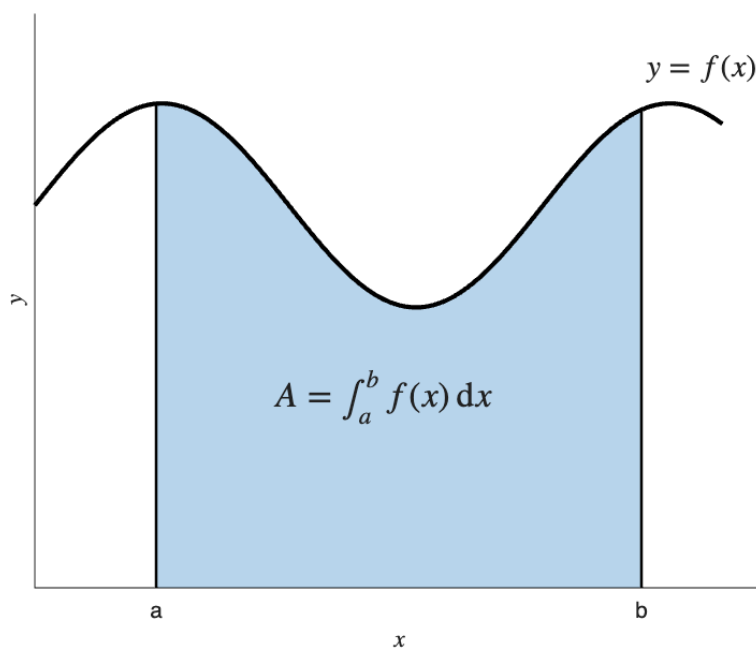
Sometimes we are interested in the value of an integral between two particular limits.

### Definition

A **definite integral** has lower and upper limits:

$$\int_a^b f(x) \, dx$$

It represents the area under the curve between  $x = a$  and  $x = b$ .



### Definition

To evaluate a definite integral:

1. Find the indefinite integral.
2. Substitute the upper limit.
3. Substitute the lower limit.
4. Subtract.

If  $F(x)$  is the integral of  $f(x)$ , then

$$\begin{aligned} \int_a^b f(x) \, dx &= [F(x)]_a^b \\ &= F(b) - F(a) \end{aligned}$$

### Worked Example

Evaluate

$$\int_1^3 2x \, dx$$

$$\begin{aligned}\int_1^3 2x \, dx &= [x^2]_1^3 \\ &= (3)^2 - (1)^2 \\ &= 9 - 1 \\ &= 8\end{aligned}$$

### Worked Example

Evaluate

$$\int_0^2 3x^4 + 1 \, dx$$

$$\begin{aligned}\int_0^2 3x^4 + 1 \, dx &= \left[ \frac{3x^5}{5} + x \right]_0^2 \\ &= \left( \frac{3(2)^5}{5} + 2 \right) - \left( \frac{3(0)^5}{5} + 0 \right) \\ &= \frac{96}{5} + 2 \\ &= \frac{106}{5} \\ &= 21.2\end{aligned}$$

## 4.2.8 Applications of Integration

### Engineering Application

A definite integral can be used to determine the area between a curve and the horizontal axis over a specified interval. This is one of the most important applications of integration and forms the basis of many engineering calculations.

### Engineering Application

If velocity is given as a function of time,  $v(t)$ , then integration can be used to calculate displacement. In other words, integration reverses the differentiation process that converts displacement into velocity.

## 4.2.9 Exercises

### Exercise

1. Evaluate the following indefinite integrals:

(a)  $\int 2x^5 \, dx$

(b)  $\int 7x^2 + 6 \sin(3x) \, dx$

(c)  $\int 4x^3 - 2x + 5 \, dx$

(d)  $\int 9e^x - \frac{5}{x^3} \, dx$

(e)  $\int 15 - 2x^{-1} + 4\sqrt{x} \, dx$

2. Evaluate the following definite integrals:

(a)  $\int_0^2 7x \, dx$

(b)  $\int_1^4 3x^2 - 7 \, dx$

(c)  $\int_{-1}^3 6e^{2x} + 6x - 1 \, dx$

(d)  $\int_1^3 \frac{5}{x} - 2x^5 \, dx$

3. Determine the area under the curve  $y = x^2$  between  $x = 0$  and  $x = 2$ .

4. A particle has velocity  $v = 6t + 2$ .

(a) Determine the displacement function,  $S$ .

(b) Given that the displacement is zero when  $t = 0$ , determine the value of the constant of integration.

## 4.2 Solutions to Exercise 4.2.9

1. Evaluate the following indefinite integrals.

(a)  $\int 2x^5 \, dx$ . Using the power rule:

$$\begin{aligned}\int 2x^5 \, dx &= \frac{2x^{5+1}}{5+1} + C \\ &= \frac{2x^6}{6} + C \\ &= \frac{x^6}{3} + C\end{aligned}$$

(b)  $\int 7x^2 + 6 \sin(3x) \, dx$ . Integrating term-by-term:

$$\begin{aligned}\int (7x^2 + 6 \sin(3x)) \, dx &= \int 7x^2 \, dx + \int 6 \sin(3x) \, dx \\ &= \frac{7x^{2+1}}{2+1} - \frac{6}{3} \cos(3x) + C \\ &= \frac{7x^3}{3} - 2 \cos(3x) + C\end{aligned}$$

(c)  $\int 4x^3 - 2x + 5 \, dx$ . Integrating term-by-term:

$$\begin{aligned}\int 4x^3 - 2x + 5 \, dx &= \int 4x^3 \, dx - \int 2x \, dx + \int 5 \, dx \\ &= \frac{4x^{3+1}}{3+1} - \frac{2x^{1+1}}{1+1} + 5x + C \\ &= x^4 - x^2 + 5x + C\end{aligned}$$

(d)  $\int 9e^x - \frac{5}{x^3} \, dx$ . First rewrite (using rules of indices) as  $\int 9e^x - 5x^{-3} \, dx$ . Therefore,

$$\begin{aligned}
\int 9e^x - \frac{5}{x^3} dx &= \int 9e^x dx - \int 5x^{-3} dx \\
&= 9e^x - \frac{5x^{-3+1}}{-3+1} + C \\
&= 9e^x - \frac{5x^{-2}}{-2} + C \\
&= 9e^x + \frac{5}{2}x^{-2} + C
\end{aligned}$$

(e)  $\int 15 - 2x^{-1} + 4\sqrt{x} dx$ . First rewrite  $\sqrt{x} = x^{1/2}$  and  $2x^{-1} = \frac{2}{x}$ . Then integrate term-by-term:

$$\begin{aligned}
\int 15 - \frac{2}{x} + 4x^{1/2} dx &= 15x - 2\ln(x) + \frac{4x^{1/2+1}}{1/2+1} + C \\
&= 15x - 2\ln(x) + \frac{4x^{3/2}}{3/2} + C \\
&= 15x - 2\ln(x) + \frac{8}{3}x^{3/2} + C
\end{aligned}$$

2. Evaluate the following definite integrals.

(a)  $\int_0^2 7x dx$ . Integrating:

$$\begin{aligned}
\int_0^2 7x dx &= \left[ \frac{7x^{1+1}}{1+1} \right]_0^2 \\
&= \left[ \frac{7x^2}{2} \right]_0^2 \\
&= \frac{7(2)^2}{2} - \frac{7(0)^2}{2} \\
&= 14
\end{aligned}$$

(b)  $\int_1^4 3x^2 - 7 dx$ . Integrating:

$$\begin{aligned}
\int_1^4 3x^2 - 7 \, dx &= [x^3 - 7x]_1^4 \\
&= (4^3 - 7(4)) - (1^3 - 7(1)) \\
&= (64 - 28) - (1 - 7) \\
&= 36 + 6 \\
&= 42
\end{aligned}$$

(c)  $\int_{-1}^3 6e^{2x} + 6x - 1 \, dx$ . Integrating:

$$\begin{aligned}
\int_{-1}^3 6e^{2x} + 6x - 1 \, dx &= \left[ \frac{6}{2} e^{2x} + 3x^2 - x \right]_{-1}^3 \\
&= [3e^{2x} + 3x^2 - x]_{-1}^3 \\
&= (3e^6 + 27 - 3) - (3e^{-2} + 3 + 1) \\
&= 3e^6 - 3e^{-2} + 20 \\
&= 1229.88 \text{ to 2 d.p.}
\end{aligned}$$

(d)  $\int_1^3 \frac{5}{x} - 2x^5 \, dx$ . Integrating:

$$\begin{aligned}
\int_1^3 \frac{5}{x} - 2x^5 \, dx &= \left[ 5 \ln(x) - \frac{2x^{5+1}}{5+1} \right]_1^3 \\
&= \left[ 5 \ln(x) - \frac{x^6}{3} \right]_1^3 \\
&= \left( 5 \ln(3) - \frac{3^6}{3} \right) - \left( 5 \ln(1) - \frac{1^6}{3} \right) \\
&= 5 \ln(3) - 243 + \frac{1}{3} \\
&= -237.17 \text{ to 2 d.p.}
\end{aligned}$$

3. Determine the area under the curve  $y = x^2$  between  $x = 0$  and  $x = 2$ . The area is given by

$$\begin{aligned} A &= \int_0^2 x^2 \, dx \\ &= \left[ \frac{x^{2+1}}{2+1} \right]_0^2 \\ &= \left[ \frac{x^3}{3} \right]_0^2 \\ &= \frac{2^3}{3} - \frac{0^3}{3} \\ &= \frac{8}{3} \end{aligned}$$

Therefore, the area under the curve is  $\frac{8}{3}$  square units.

4. A particle has velocity  $v = 6t + 2$ .

- (a) Determine the displacement function,  $S$ . Since

$$v = \frac{dS}{dt},$$

the displacement is obtained by integration:

$$\begin{aligned} S &= \int 6t + 2 \, dt \\ &= \int 6t \, dt + \int 2 \, dt \\ &= \frac{6t^{1+1}}{1+1} + 2t + C \\ &= \frac{6t^2}{2} + 2t + C \\ &= 3t^2 + 2t + C \end{aligned}$$

Therefore,

$$S = 3t^2 + 2t + C$$

- (b) Given that the displacement is zero when  $t = 0$ , determine the value of the constant of integration. Substituting  $S = 0$ ,  $t = 0$  into  $S = 3t^2 + 2t + C$ , gives

$$0 = 3(0)^2 + 2(0) + C$$

$$0 = C$$

Therefore,  $C = 0$ . The displacement function is therefore

$$S = 3t^2 + 2t$$